Fine-Tuning DETR for Maritime Object Detection

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Problem and Aim

Problem

- Maritime images contain multiple objects such as ships, boats, buoys, sail boats, and kayaks.
- Reliable object detection is required for real-world maritime applications such as:
  - unmanned surface vessels (USVs)
  - port monitoring
  - collision avoidance.
- Challenging due to small/distant objects, overlapping vessels, poor visibility, and complex backgrounds.

Aim

- Fine-tune a pretrained DETR ResNet-50 model for maritime object detection.
- Adapt the model from generic COCO object detection to maritime-specific classes.
- Evaluate whether fine-tuning improves detection performance on unseen maritime images.

Dataset Selection and Justification

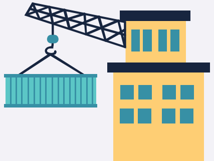
Singapore Maritime Dataset

- Maritime image dataset collected from Roboflow Universe.
- Uses COCO annotation format, including:
  - Object class labels
  - Bounding box coordinates
  - Image metadata
- Suitable for this project because it has real-world images with objects such as:
  - Vessel-ship
  - Ferry
  - Speed boat
  - Buoy
  - Sail boat
  - Kayak

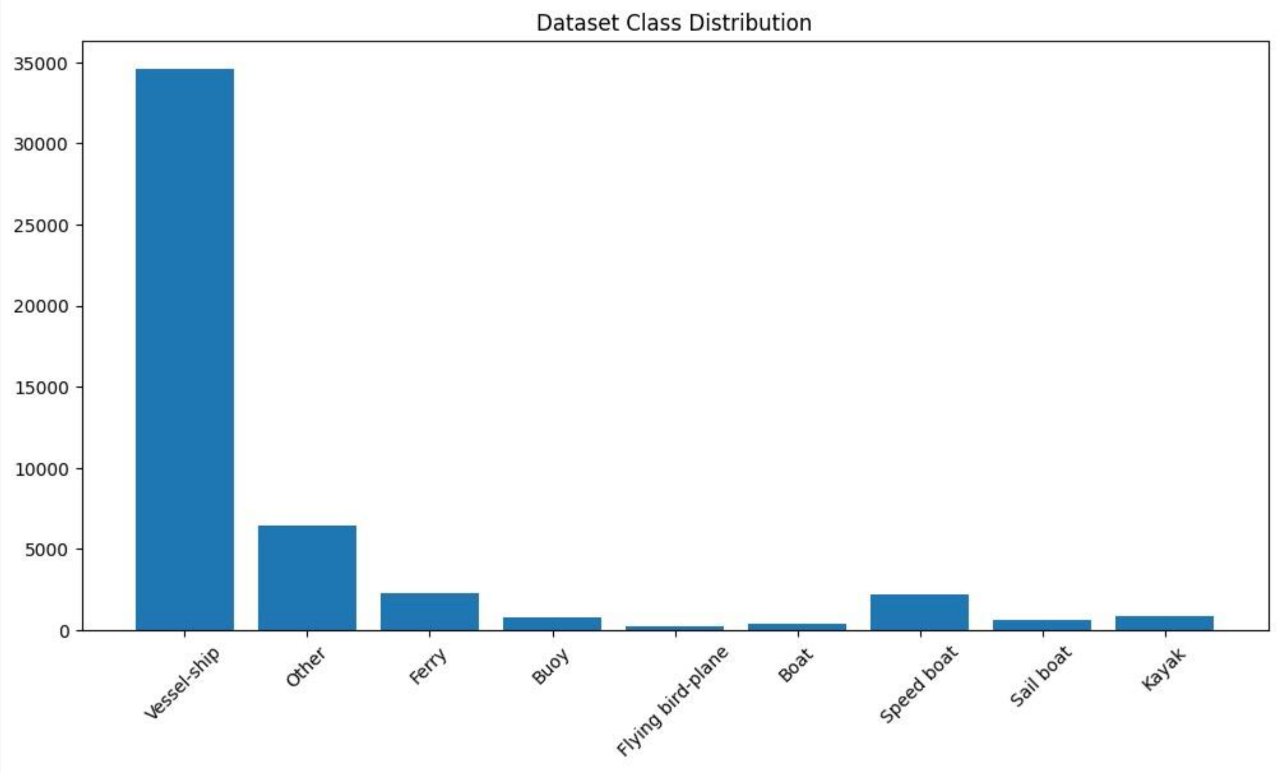
Property	Value
Annotation Format	COCO JSON
Number of Images	5446
Number of Annotations	41747
Number of Classes	10
Training Images	4356 (80%)
Validation Images	545 (10%)
Testing Images	545 (10%)

Dataset Example with COCO Annotations

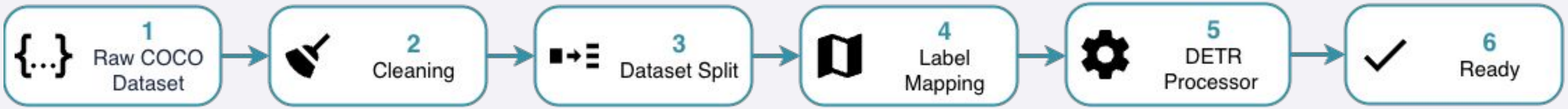
COCO annotation = class label + bounding box [x, y, width, height]



Class Distribution and Imbalance



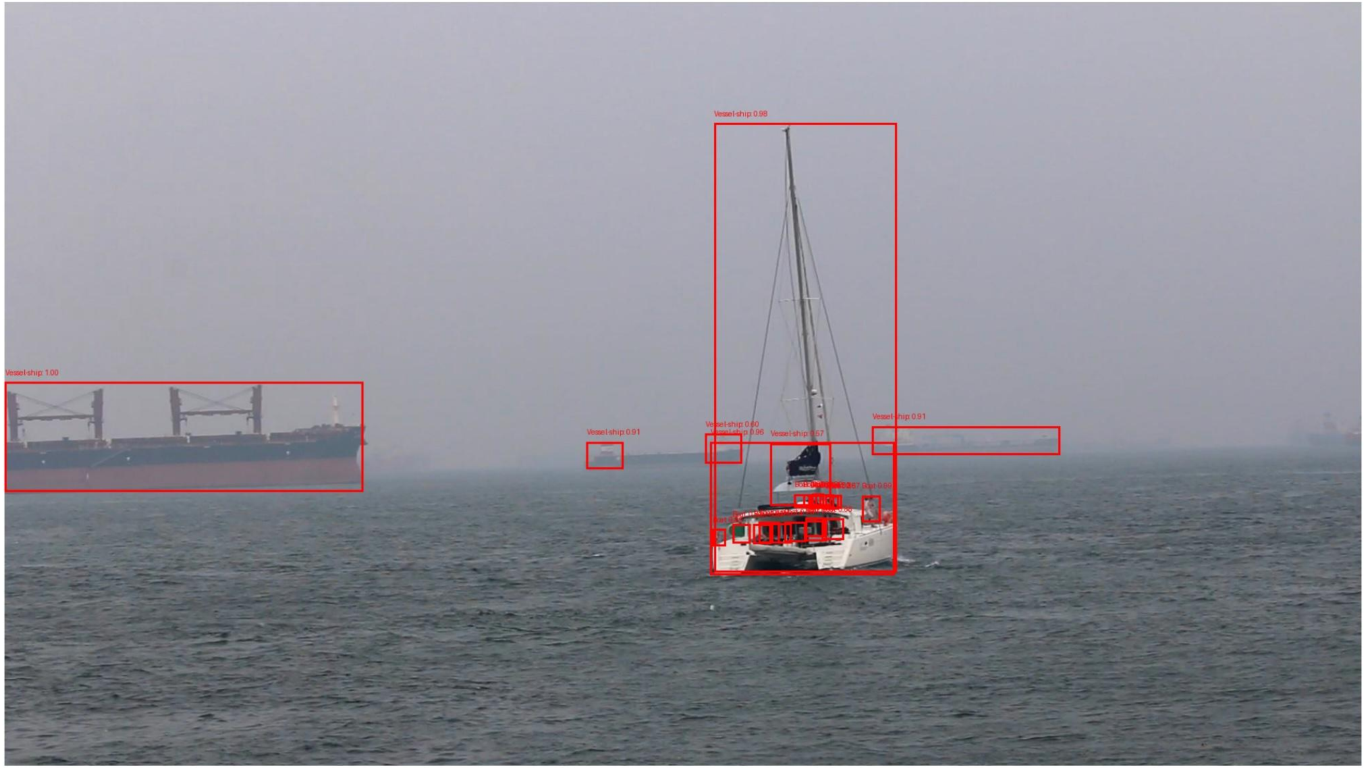
Preprocessing Strategy



1. Removed missing image records and invalid annotations.
2. Verified bounding box parser was loading boxes correctly on sample images.
3. Split dataset into:
   a. 80% training
   b. 10% validation
   c. 10% testing
4. Created dictionaries for *id2label* and *label2id* mappings for the maritime classes.
5. Used DetrImageProcessor to resize, normalize and format image/bounding boxes for DETR.

Baseline DETR Performance

Baseline DETR Predictions Before Fine-Tuning

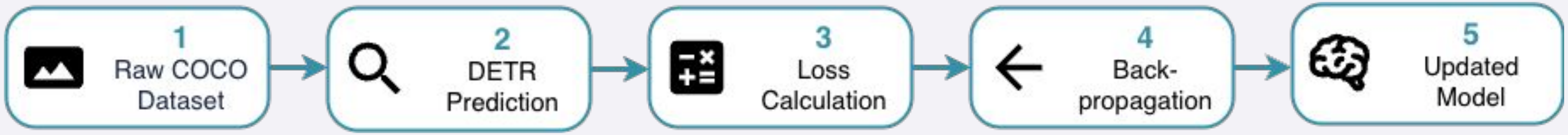


Model Selection and Configuration

DETR ResNet-50

- Used a pretrained facebook/detr-resnet-50 model from Hugging Face.
- DEtection TRansformer (DETR) was selected because:
  - It is designed for object detection and can detect multiple objects in one image.
  - Transformer attention helps analyse complex scenes.
  - It works seamlessly with COCO format annotations.
- The model combines:
  - A ResNet-50 backbone for visual feature extraction.
  - A Transformer encoder-decoder for global object reasoning.
  - Prediction heads for object class labels and bounding boxes.
- The pretrained Facebook model was trained on the COCO dataset, which contains 91 categories.
  - Transfer learning allowed the model to reuse pretrained visual features while adapting to maritime-specific objects.

Training and Fine-Tuning Process



1. Started with pretrained DETR ResNet-50 weights.
2. Replaced the original COCO classification layer to match the maritime dataset classes.
3. Image and COCO annotations were passed through the model during training.
4. DETR calculated a combined detection loss:
    a. Classification loss.
    b. Bounding box regression loss.
5. Backpropagation was used to update the model weights.
6. AdamW optimizer adjusted the model parameters after each batch.
7. Validation loss was tracked after each epoch to select the best model.

Experiment Configurations

Experiment	Learning Rate	Weight Decay	Epochs
Experiment 1	1×10^{-5}	0.0001	5
Experiment 2	1×10^{-5}	0.0001	10
Experiment 3	5×10^{-5}	0.0001	5

Experiment 1: Baseline fine-tuning setup using a conservative learning rate and 5 epochs.

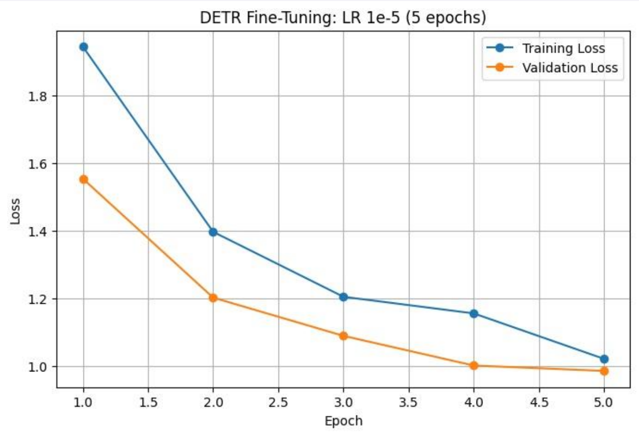
Experiment 2: Same learning rate, but trained for 10 epochs to test whether longer training improved convergence.

Experiment 3: Higher learning rate to test whether faster optimisation improved performance or made training unstable.

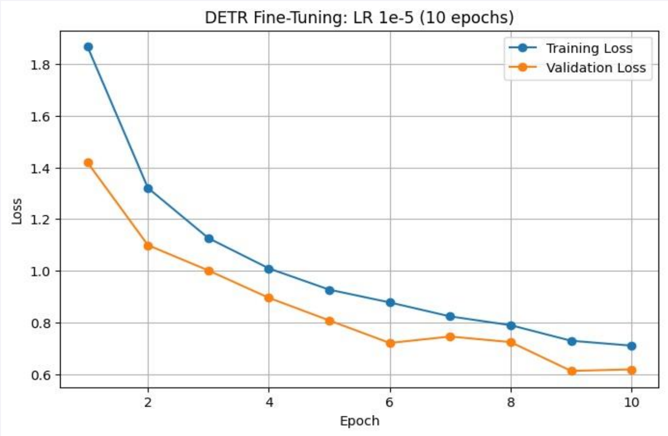
Constant Setting: Weight decay was kept the same for all experiments to make the comparison fair.

Training
Results

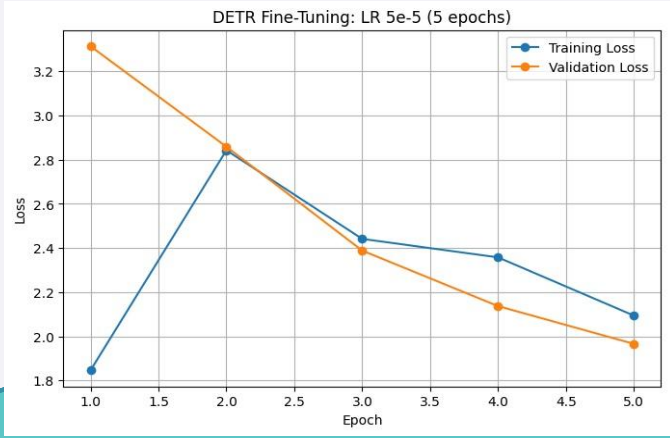
Experiment 1



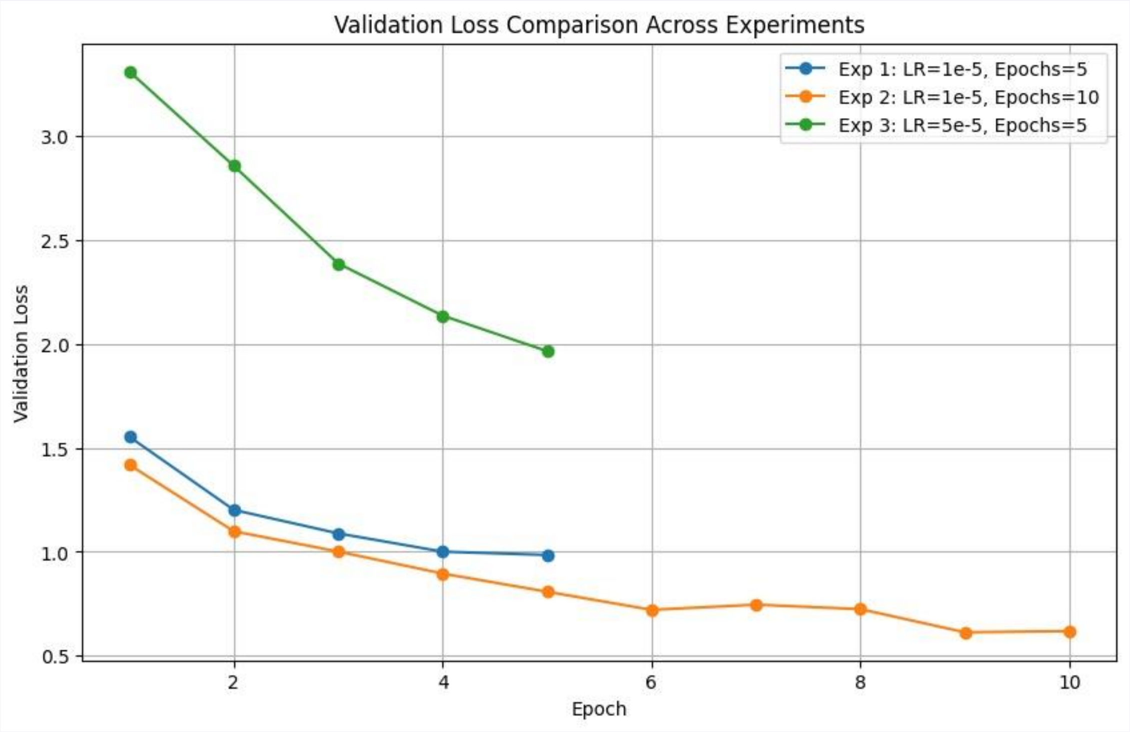
Experiment 2



Experiment 3



Combined Validation Loss



Experiment 1: Stable learning, but limited by 5 epochs.

Experiment 2: Lowest validation loss and best convergence.

Experiment 3: Higher learning rate caused unstable training.

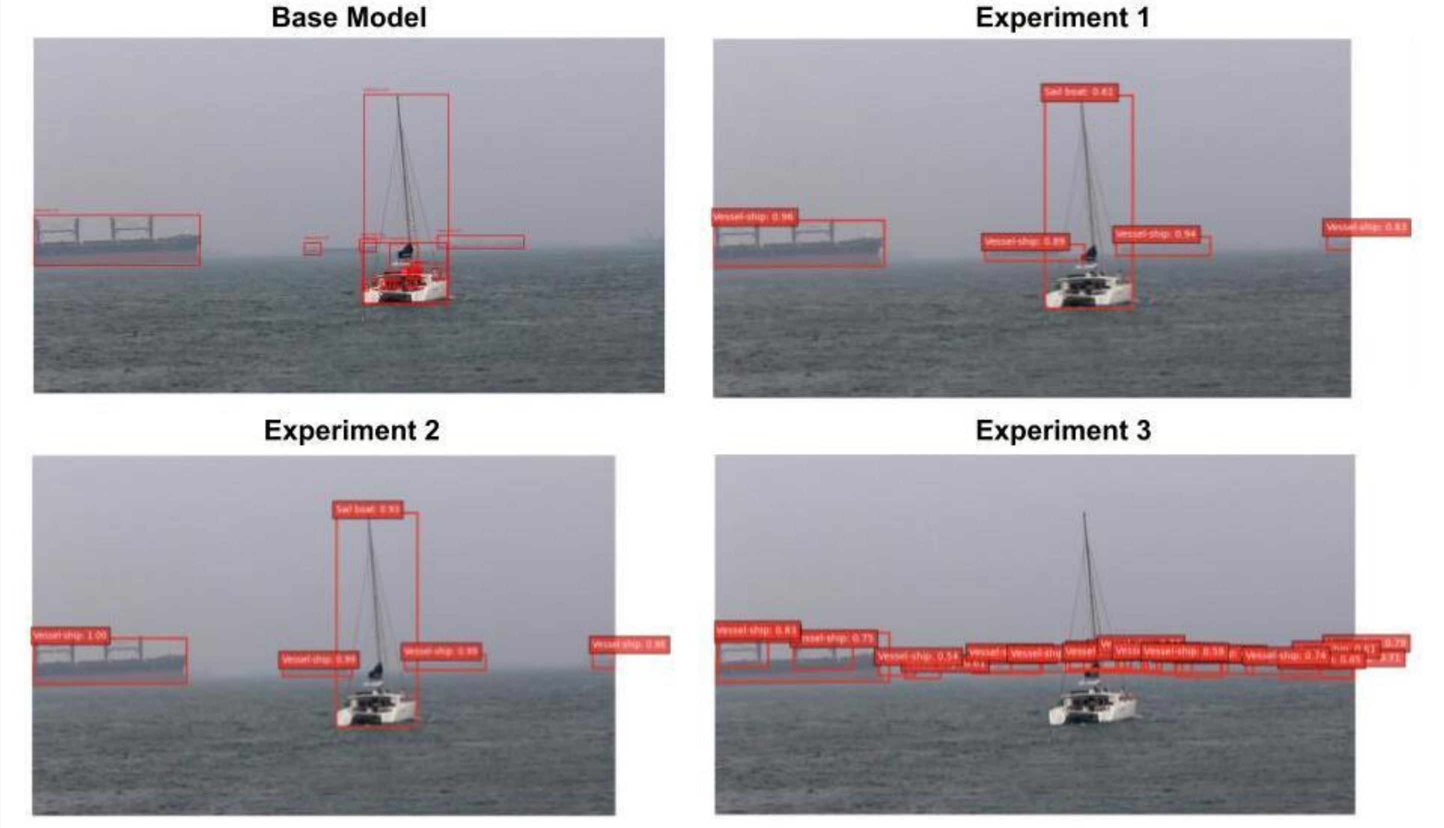
Best Model: Experiment 2 selected for final evaluation.

Final Evaluation Metrics

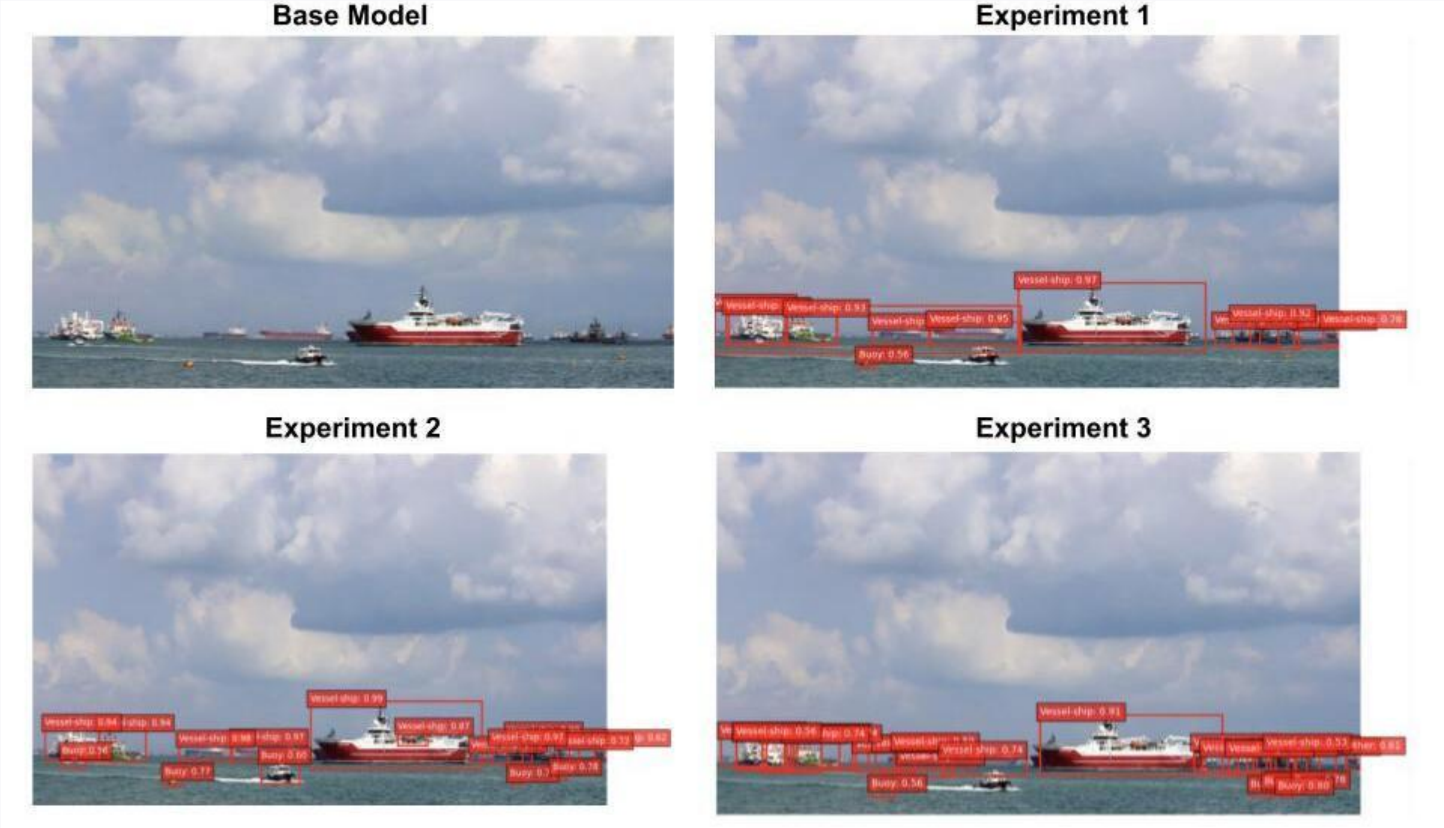
Experiment	mAP	mAP@50	mAP@75	Mean Recall
Experiment 1	0.2181	0.3558	0.2131	0.3451
Experiment 2	0.3491	0.5443	0.3601	0.4634
Experiment 3	0.0285	0.0713	0.0185	0.0108

Experiment 2 achieved the best performance across all metrics, showing that longer training with a conservative learning rate created the strongest detection model.

Qualitative Results 1



Qualitative Results 2



Ethical and Societal Implications

- Maritime object detection can support safety, surveillance, port monitoring and autonomous navigation.
- False negative can create safety risks if vessels or objects are missed.
- False positives may cause unnecessary alerts or poor operational decisions.
- Dataset imbalance may make the model more reliable for common vessel types rather than rare objects.
- The system should **support** human decision-making, **not replace** oversight.

Conclusion

- A pretrained DETR ResNet-50 model was fine-tuned for maritime object detection.
- Experiment 2 achieved the best overall performance using a learning rate of 1×10^{-5} for 10 epochs.
- Fine-tuning improved detection compared with the baseline pretrained model.
- Remaining limitations include class imbalance, small object detection and bounding box localisation.
- Future improvements could include data augmentation, class balancing, and longer training with early stopping.

Thanks!

Any questions?

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